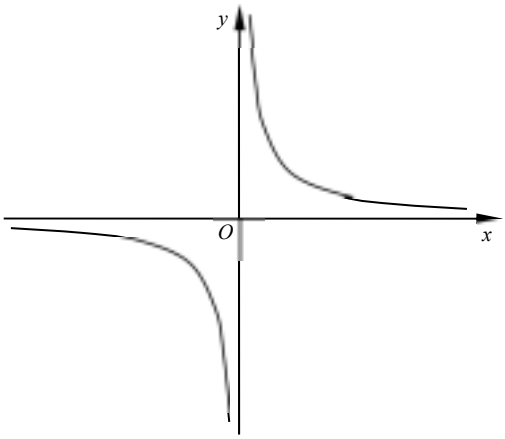
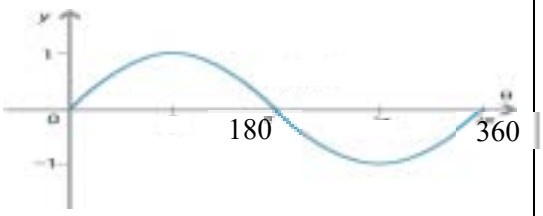
Question	Answer	Marks	Partial Marks
7(b)(i)	Correct sketch 	**2**	**B1** for one branch correct or an attempt at the correct shape Maximum 1 mark if sketch crosses x-axis or y-axis
7(b)(ii)	$\pm\frac{1}{2}$ **nfww**	**2**	**M1** for $4x^2 = 1$ oe or **B1** for $\frac{1}{2}$ or $-\frac{1}{2}$ nfww
7(c)(i)	Correct sketch through (0, 0) (180, 0) and (360, 0) with max and min at 1 and –1 resp. 	**2**	**B1** for correct sine curve shape, starting at the origin, with minimum of 1 cycle.
7(c)(ii)	199.5 or 199.47... and 340.5...	**3**	**B2** for one correct or **M1** for $\sin x = -\frac{1}{3}$ oe If 0 scored, **SC1** for two reflex angles with a sum of 540 or 2 non-reflex angles with a sum of 180
8(a)	$4x + 3(x + 27) = 194.75$ or $4x + 3x + 81 = 194.75$	**M1**	
	16.25 cao	**B2**	**M1** for $7x = k$ where $k < 194.75$ or **B1** for answer 16.3
8(b)	$x^2 - 20x - 69\,[=0]$ oe or $y^2 + 116y - 861\,[=0]$ oe	**M2**	**M1** for $x^2 + 4(-8 - 5x) = 37$ oe or for $37 - 4y = \left(\frac{-8-y}{5}\right)^2$ oe or for $x^2 + 4y = 37$ and $20x + 4y = -32$ subtracted with no more than one error

Question	Answer	Marks	Partial Marks
	$(x + 3)(x – 23)$ [= 0] oe or $(y – 7)(y + 123)$ [= 0] oe	**M1**	correct method to solve *their* quadratic e.g. $x = \dfrac{-(-20) \pm \sqrt{(-20)^2 - 4 \times 1 \times (-69)}}{2 \times 1}$ or $x – 10 = \pm 13$ or $x – 10 = \pm\sqrt{169}$
	$x = -3$ $y = 7$ $x = 23$ $y = -123$ final answer	**B2**	**B1** for one correct pair or two correct x values or two correct y values
8(c)	$2\pi x \times 6x + 2\pi x^2$ or $2\pi x(6x + x)$	**M2**	or **M1** for $2\pi x \times 6x$ or $2\pi x^2$
	Their $(2\pi x \times 6x + 2\pi x^2) = 4\pi r^2$	**M1**	Dep on at least on M1 earned *Their* LHS must be an area in terms of x only
	At least one further stage of working leading to $r^2 = \dfrac{7}{2}x^2$	**A1**	with no error seen
9(a)(i)	311 or 311.0 to 311.1	**3**	**M2** for $11 \times 11 + 2 \times \dfrac{1}{4} \times \pi \times 11^2$ oe or **M1** for [2 ×] $\dfrac{1}{4} \times \pi \times 11^2$ or 11×11 oe
9(a)(ii)	78.6 or 78.55 to 78.56...	**3**	**M2** for $4 \times 11 + 2 \times \dfrac{1}{4} \times 2 \times \pi \times 11$ oe or **M1** for [2 ×] $\dfrac{1}{4} \times 2 \times \pi \times 11$ or 4×11 oe
9(b)	35.2 or 35.3 or 35.239… to 35.28	**4**	**M3** for [tan =] $\dfrac{7}{\sqrt{7^2 + 7^2}}$ or [sin =] $\dfrac{7}{\sqrt{7^2 + 7^2 + 7^2}}$ or [cos =] $\dfrac{\sqrt{7^2 + 7^2}}{\sqrt{7^2 + 7^2 + 7^2}}$ OR **M2** for $AG = \sqrt{7^2 + 7^2 + 7^2}$ or for $\sqrt{7^2 + \left(\dfrac{7}{\sin 45}\right)^2}$ oe or for $AC = \sqrt{7^2 + 7^2}$ or $\dfrac{7}{\sin 45}$ oe OR **M1** for $7^2 + 7^2$ or for implicit trigonometry or identifying correct angle